Collision lab

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Test momentum conservation with measured before-and-after velocities.

SUCCESS CRITERIA - I can...

1. Record masses, signs and velocities consistently.
2. Compare total momentum before and after.
3. Discuss external impulse and measurement variation.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “trial” and “momentum” to explain your model or result. Sentence starters are optional.

trial: one run or repetition of a test

momentum: mass times velocity, including direction

external: acting from outside the chosen system

01 Warm-up: bring back an earlier idea

A 3 kg object slows from 7 to 3 m/s in 1 s. **Find** impulse and average net force. Repeat the force calculation for the same velocity change over 2 s.

Word help: impulse = net force acting over time; change in momentum

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $p_{total} = m_1v_1 + m_2v_2$; $\Delta p_{total} = p_{total,\,f} - p_{total,\,i}$. Compare just before/after collision. Use approved carts; record actual trials separately from practice data.

Worked reasoning: Supplied trials give total momenta 6.0 before and 5.8 kg·m/s after. The change is -0.2 kg·m/s. Compare this with measurement uncertainty before attributing it to a new physical effect.

Your turn: Draw a before-and-after momentum comparison and name one possible external interaction.

Sentence starter: The ... in my model represents ... . Its label shows ... .

Before	After

Label mass, velocity and momentum.

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Read and interpret the model

Supplied case: cart A is 0.50 kg at +0.40 m/s; B is 0.50 kg at rest. After sticking, both move at +0.20 m/s.
Find total momentum before and after.

Word help: at rest = not moving in the chosen reference frame

Sentence starter: I found … [unit]. This value means … in this situation.

Cart	Before (m/s)	After (m/s)
A: 0.50 kg	+0.40	+0.20
B: 0.50 kg	0	+0.20

Marzano check - circle: 1 2 3

04 Apply the physics

In a lab collision, a 0.50 kg cart moving at +0.60 m/s sticks to a stationary 0.75 kg cart. The measured common final velocity is +0.23 m/s. Complete the table, find total momentum before and after, and calculate the percent deviation relative to the initial momentum.

Sentence starter: I found … [unit]. This value means … in this situation.

Cart	Mass (kg)	v before (m/s)	v after (m/s)
A	0.50	+0.60	+0.23
B	0.75	0	+0.23

Marzano check - circle: 1 2 3

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Explain your reasoning

Design a table for three trials with masses and signed velocities before/after.
Name one possible external impulse and one measurement limitation.

Sentence starter: My design shows … . It meets the requirement by … .

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

Launch at 40 m/s from level ground at 30° or 60°.
Use g=10 and no drag.
Compare ranges and flight times.
Label separate horizontal and vertical motion.

Sentence starter: Both … . They differ in … because … .



Add velocity / acceleration arrows.

Marzano check - circle: 1 2 3

07 Exit check

A nonzero initial momentum is 0.40 kg·m/s and final is 0.38.
Find percent deviation and state why this is not automatically a 2% uncertainty.

Sentence starter: Under the assumption …, the evidence … supports … because … .

Object / trial	Mass	Velocity	Momentum
A	____	____	____
B	____	____	____

Marzano check - circle: 1 2 3

08 STRETCH: test the model

Initial total is 2.0±0.1 and final total 1.8±0.1 kg·m/s.
Find worst-case bounds for the change. If contact lasts 0.2 s, bound average external force.

Sentence starter: I found … [unit]. This value means … in this situation.

Object / trial	Mass	Velocity	Momentum
A	____	____	____
B	____	____	____

Marzano check - circle: 1 2 3